



Cell Biology

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Lecture: 2(Lg)

Eukaryotic cells

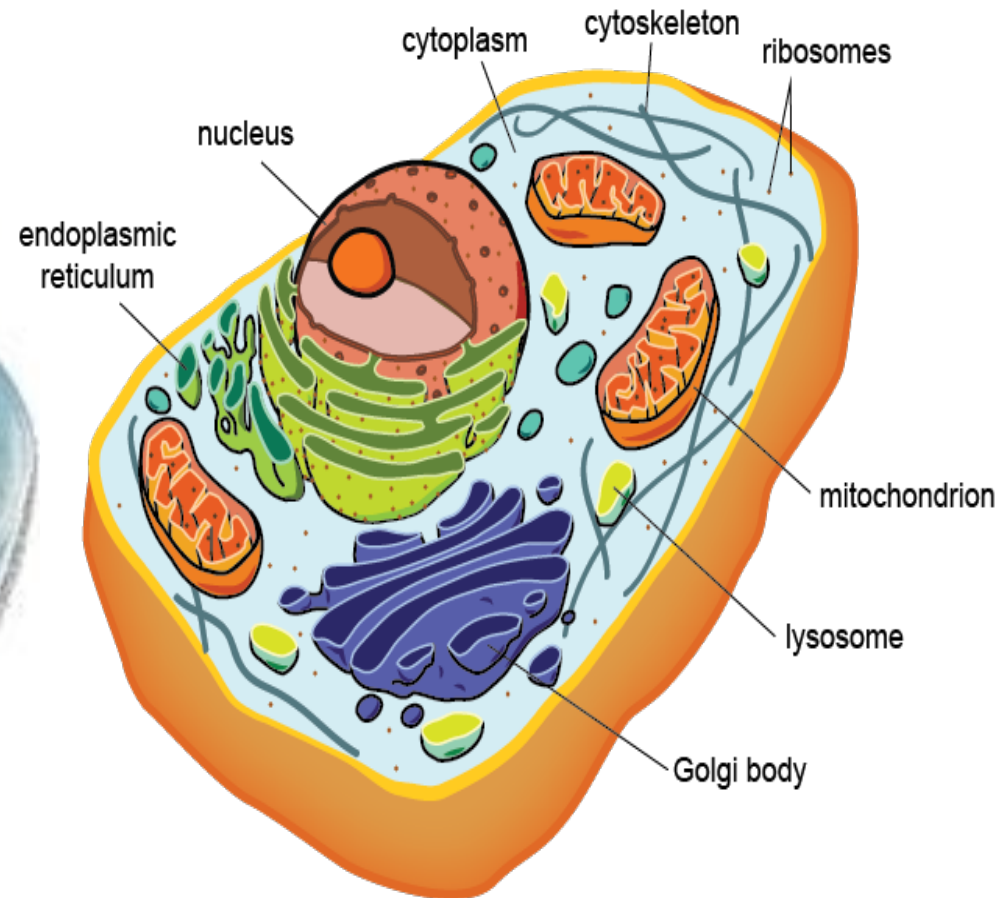
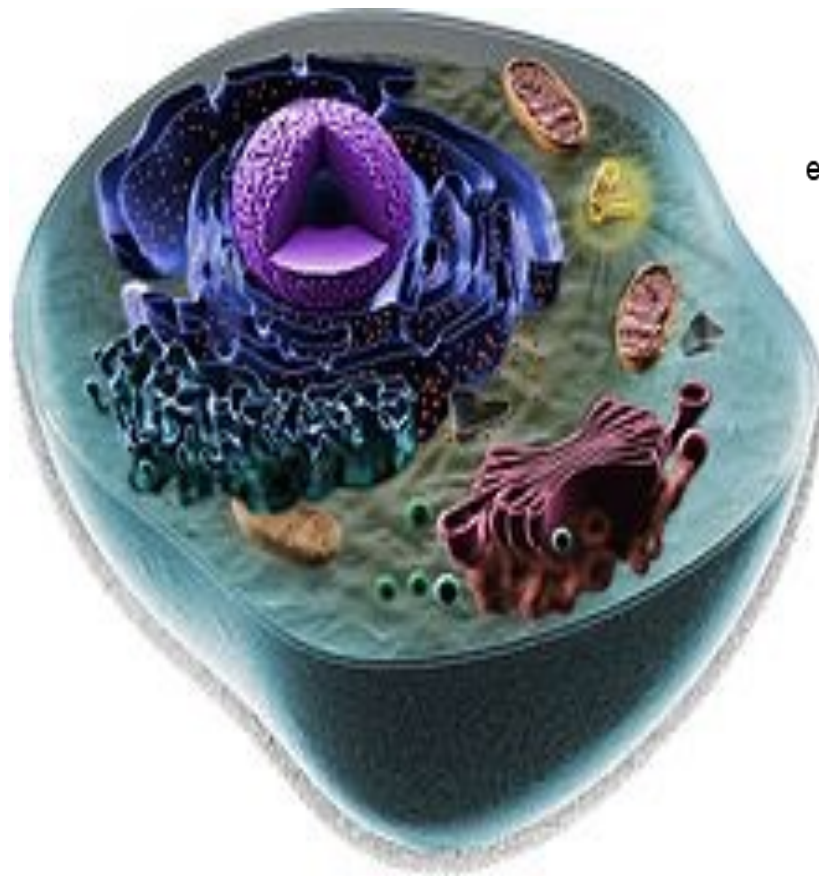
A eukaryote is any organism whose cells contain a nucleus and other organelles enclosed within membranes. The defining feature that sets eukaryotic cells apart from prokaryotic cells is that they have membrane-bound organelles, especially the nucleus, which contains the genetic material and is enclosed by the nuclear envelope. The presence of a nucleus gives eukaryotes their name, which comes from the Greek (eu, "well" or "true") and (karyon, "nut" or "kernel").

*Eukaryotic cells also contain other membrane-bound organelles such as mitochondria and the Golgi apparatus. In addition, plants and algae contain chloroplasts.

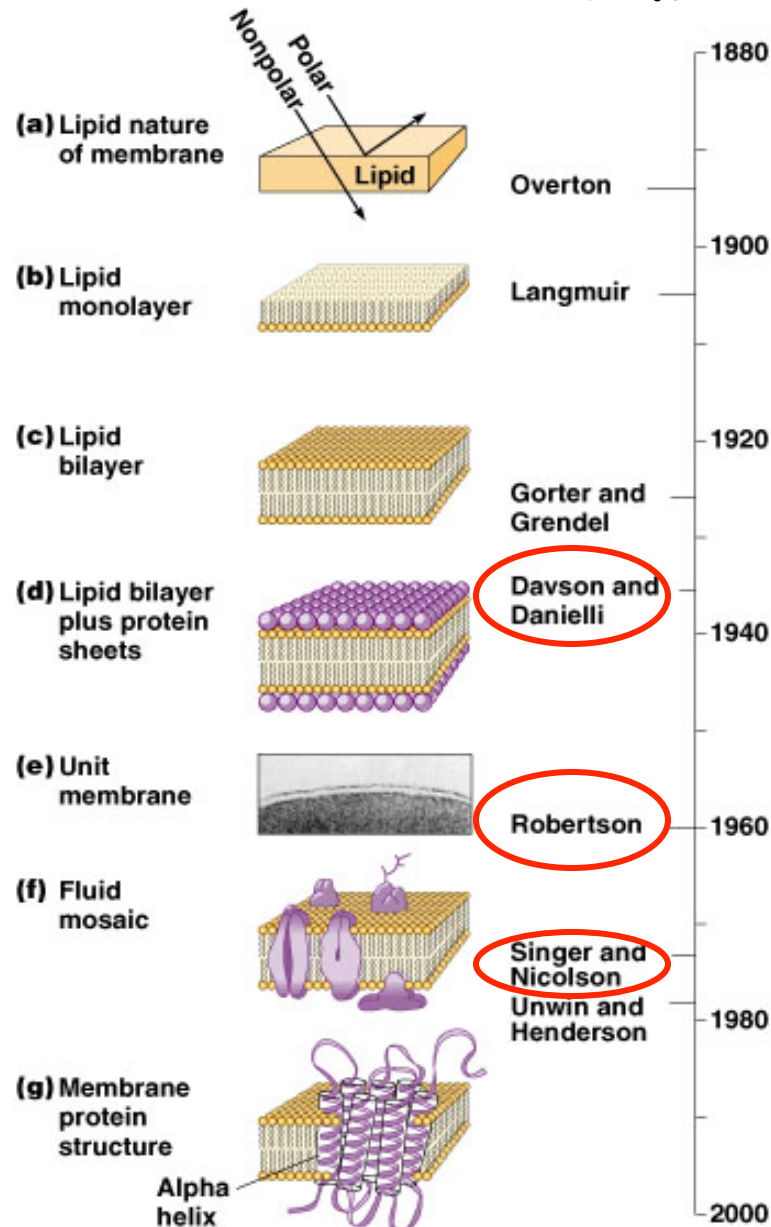
Eukaryotic organisms may be unicellular or multicellular. Only eukaryotes form multicellular organisms consisting of many kinds of tissue made up of different cell types.

*Eukaryotes can reproduce both asexually through mitosis and sexually through meiosis and gamete fusion.

Eukaryotic cell



Membrane-models



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Lipid soluble substances enter the cell rapidly

Benzine-lipid mixture, after evaporation of benzine, a **molecular lipid film** is formed

Benzine-soluble lipids form a **bilayer** film on the surface of water

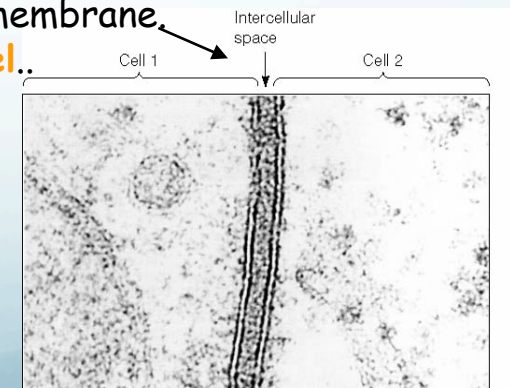
Proteins are integral parts of the cell membrane. The lipid bilayer is covered by a layer of proteins inside and outside. Partially explains the rapid passage of proteins, sugars, ions and other hydrophilic substances.

Discovery of electron microscope. The cells are covered with a plasma membrane.

"Unit-membrane" model..

Mosaic arrangement of proteins in the membrane.

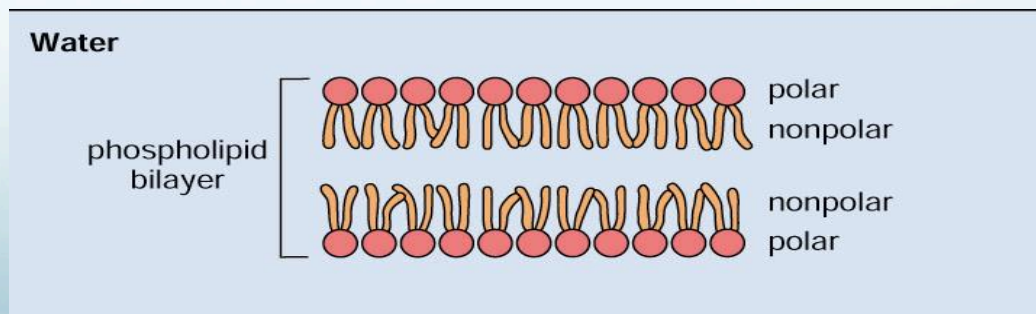
Some proteins cross the membrane, **transmembrane proteins**

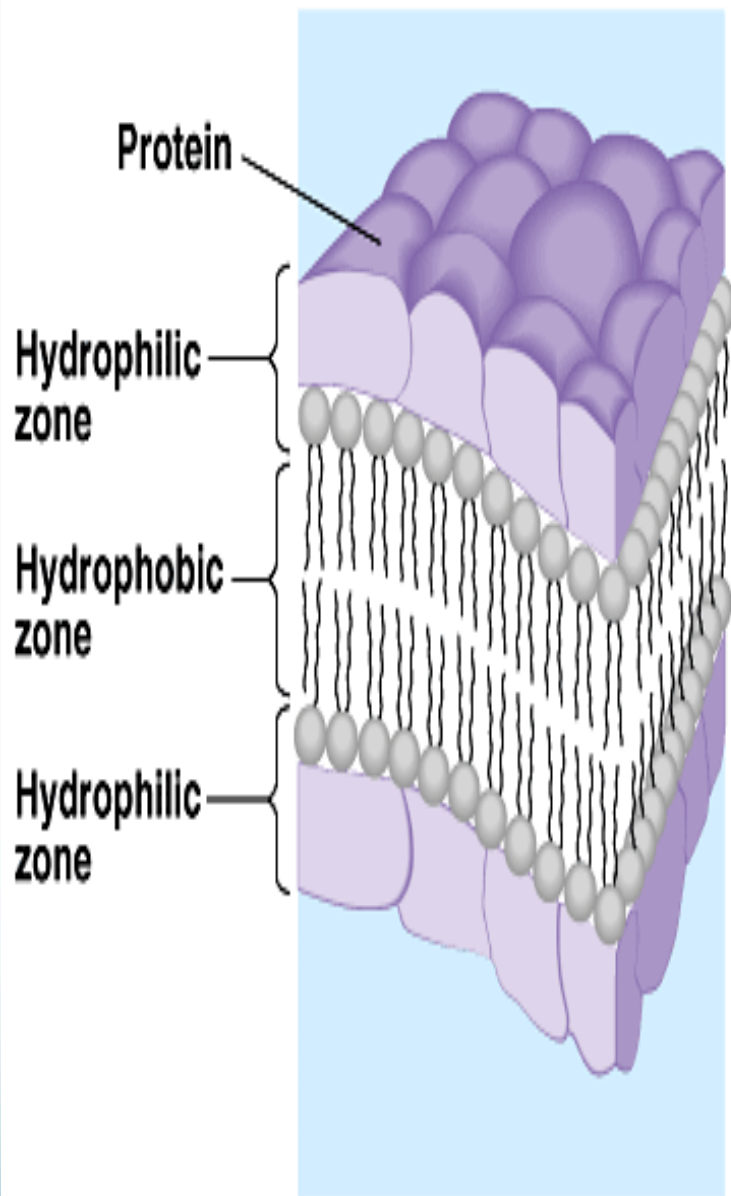


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History of the Membrane Idea

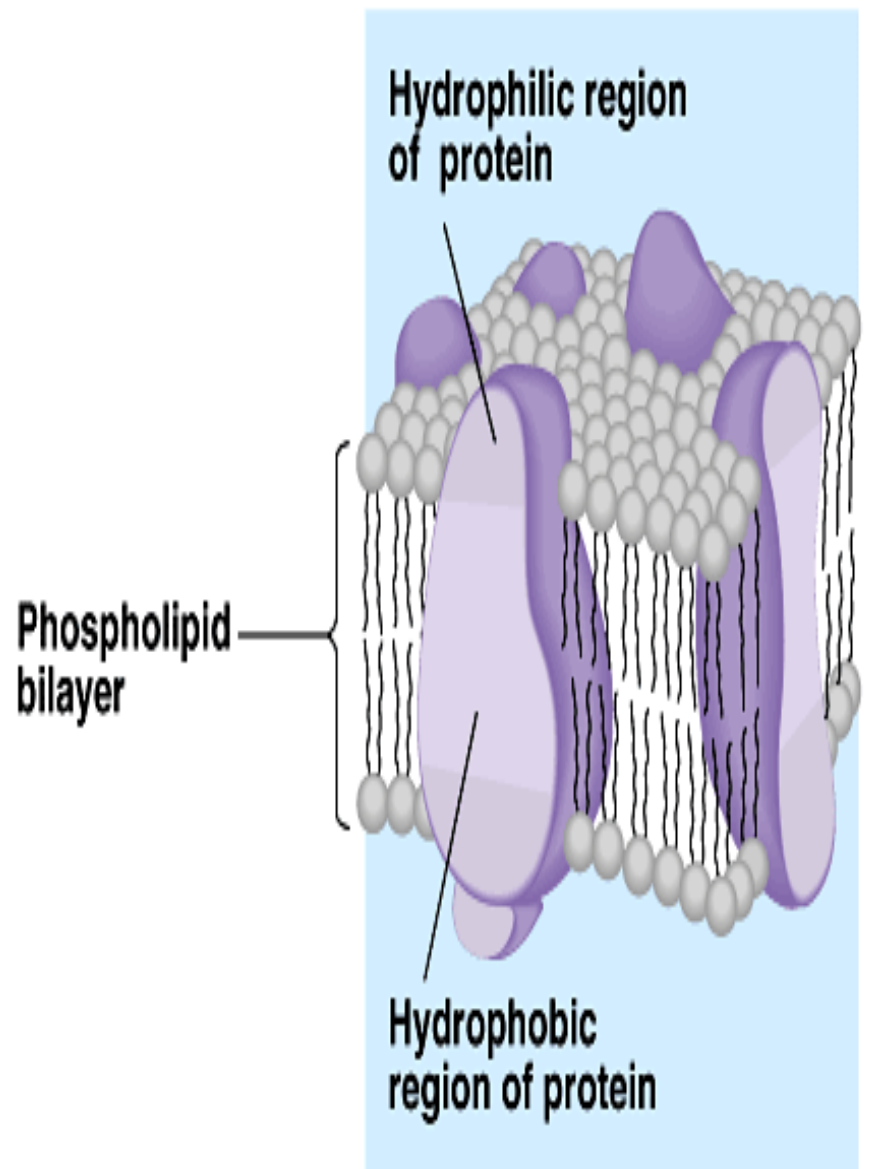
- **1925— Gorter and Grendel—proposed lipid bilayer model of cell membrane •**
- **1940s— Daniel and Davson— proposed a sandwich model: protein, phospholipid, and protein.**
- **1972— Singer and Nicholson—introduced the fluid mosaic model.**





sandwich model

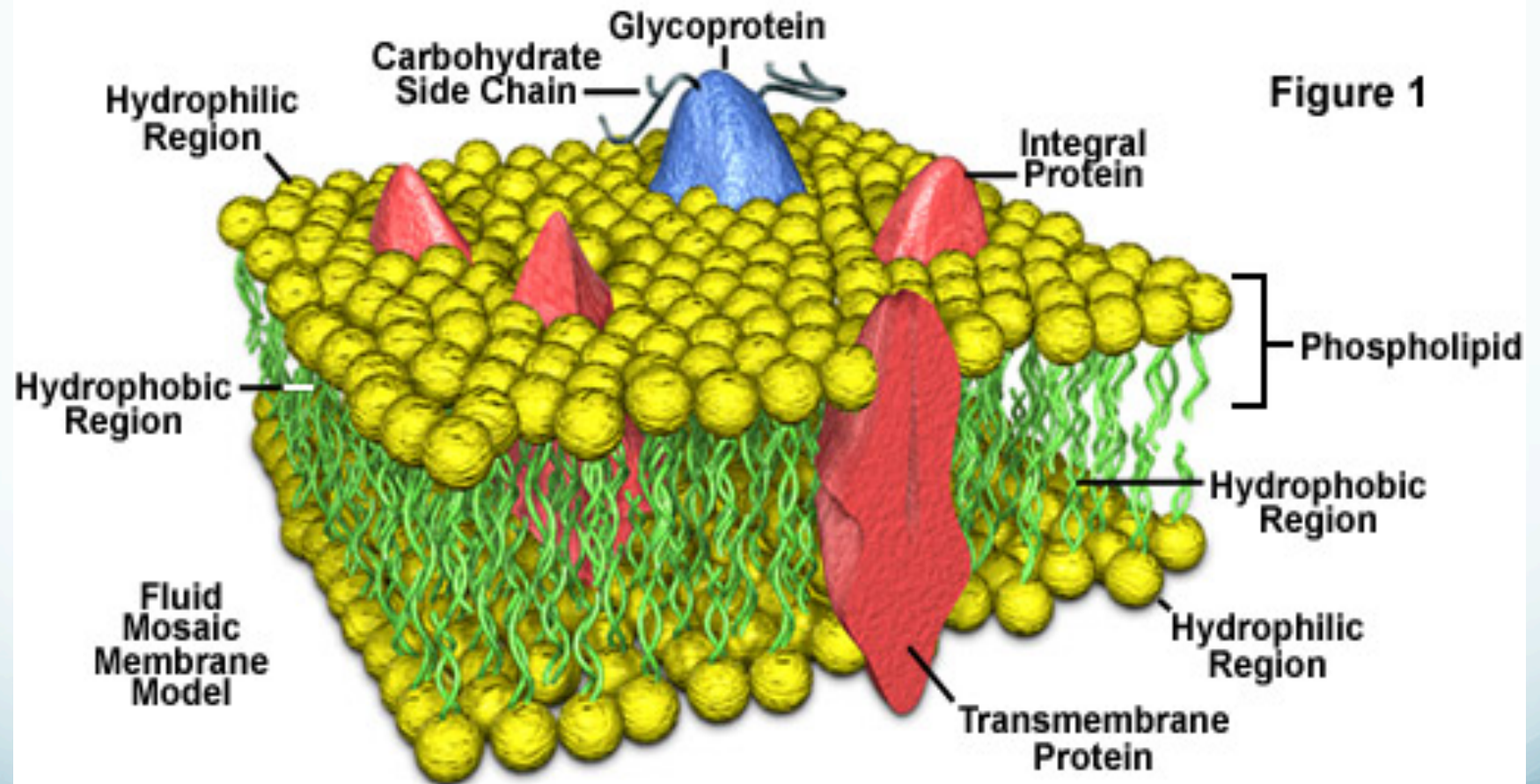
(a) The Davson-Danielli model



Singer and Nicholson model

(b) Current fluid mosaic model

Plasma Membrane Structural Components



Plasma membrane

Introduction

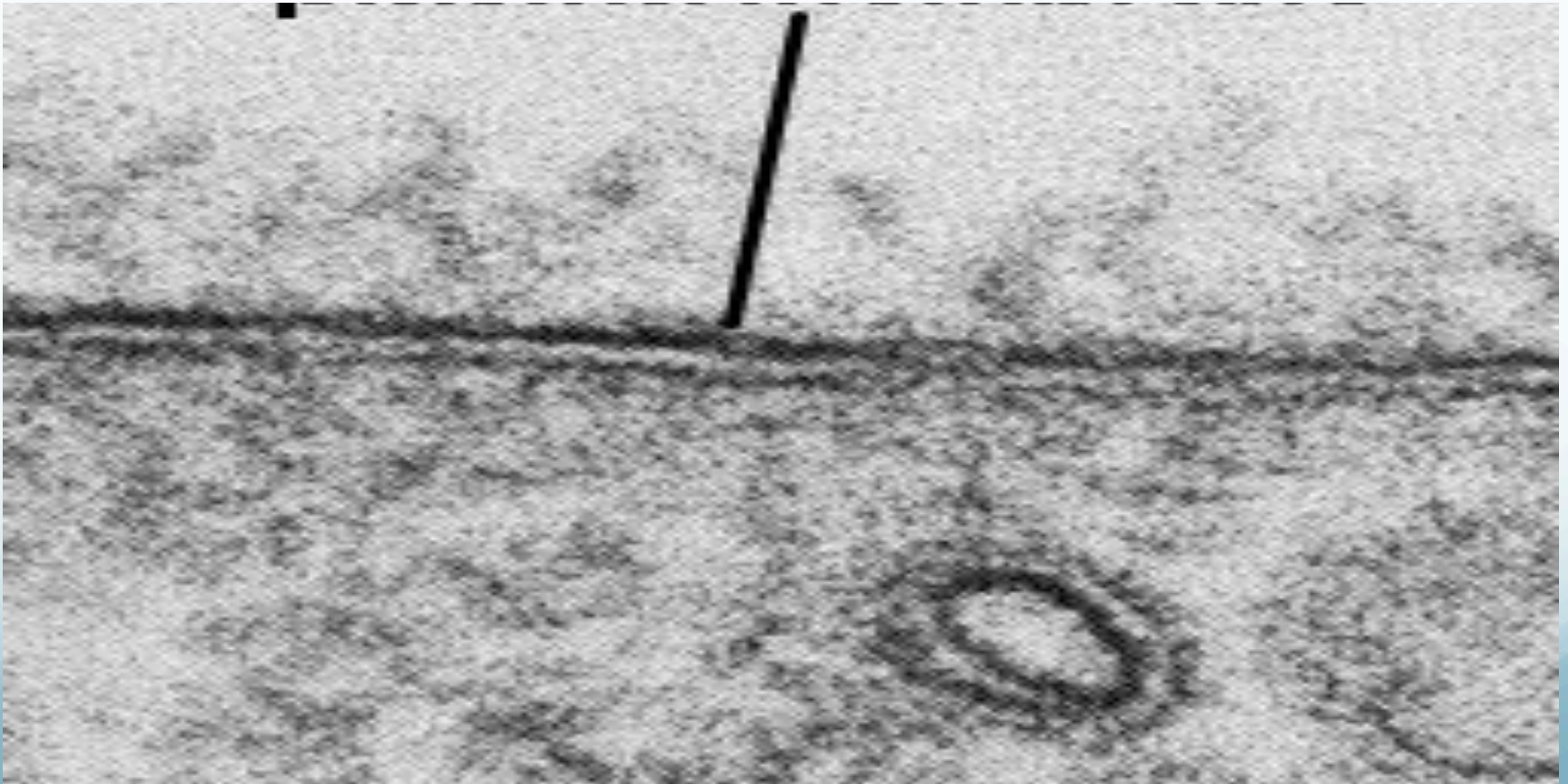
- All eukaryotic cells are enveloped by a limited membrane composed of phospholipids, cholesterol, protein and oligosaccharide.
The plasma membrane acts as a selective barrier that regulate the passage of certain materials into and out of the cell.
- In addition of this, the membrane also carry out number of specific recognition and regulatory functions

Plasma membrane

Introduction

- By T.E.M. the plasma membrane is composed of three layers
 - *An outer electron –dense layer
 - *An inner electron –dense layer.
 - *An intermediate, electron lucent layer.
- The total thickness of plasma membrane is about (7.5)nm Thick and appears as tri laminar structure this three layer structure has been known as the (Unit membrane)

Plasma membrane

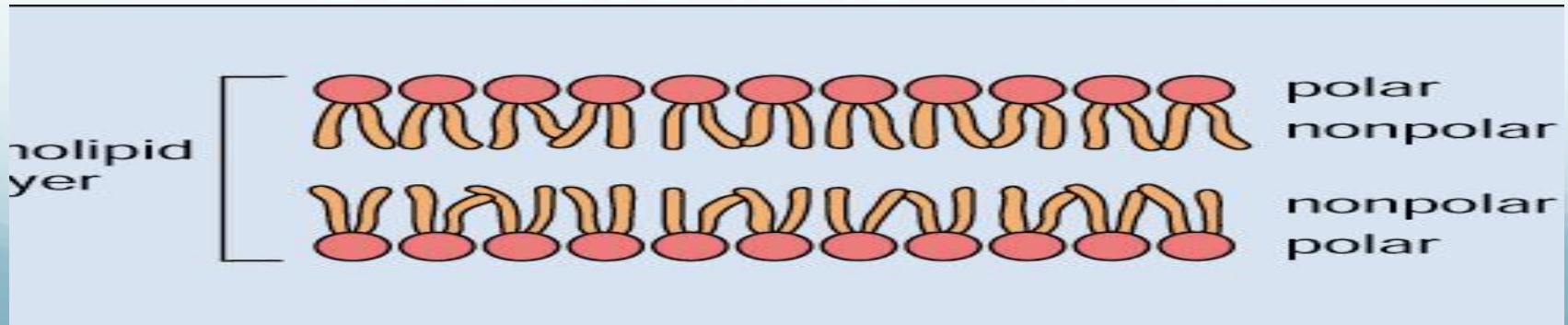


Molecular composition of plasma membrane

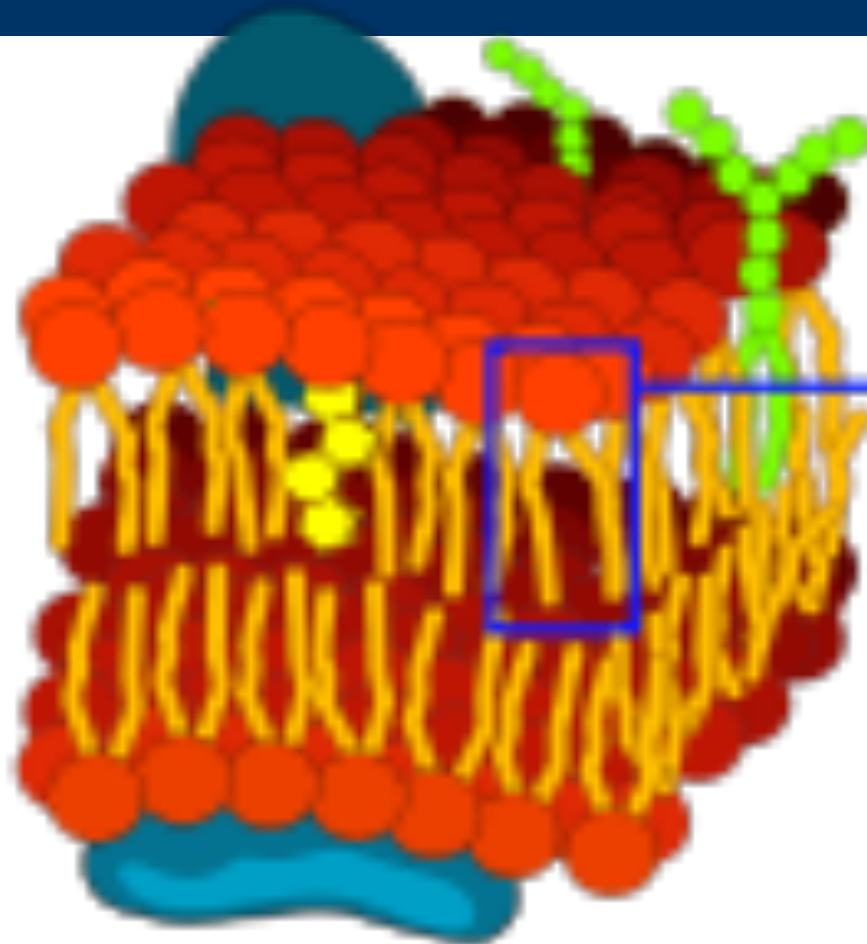
- Plasma membrane of eukaryotic cells is composed of
- 1-Lipids (phospholipids & cholesterol)
- 2-Proteins (integral & Peripheral)
- 3-Carbohydrate.

1-lipids

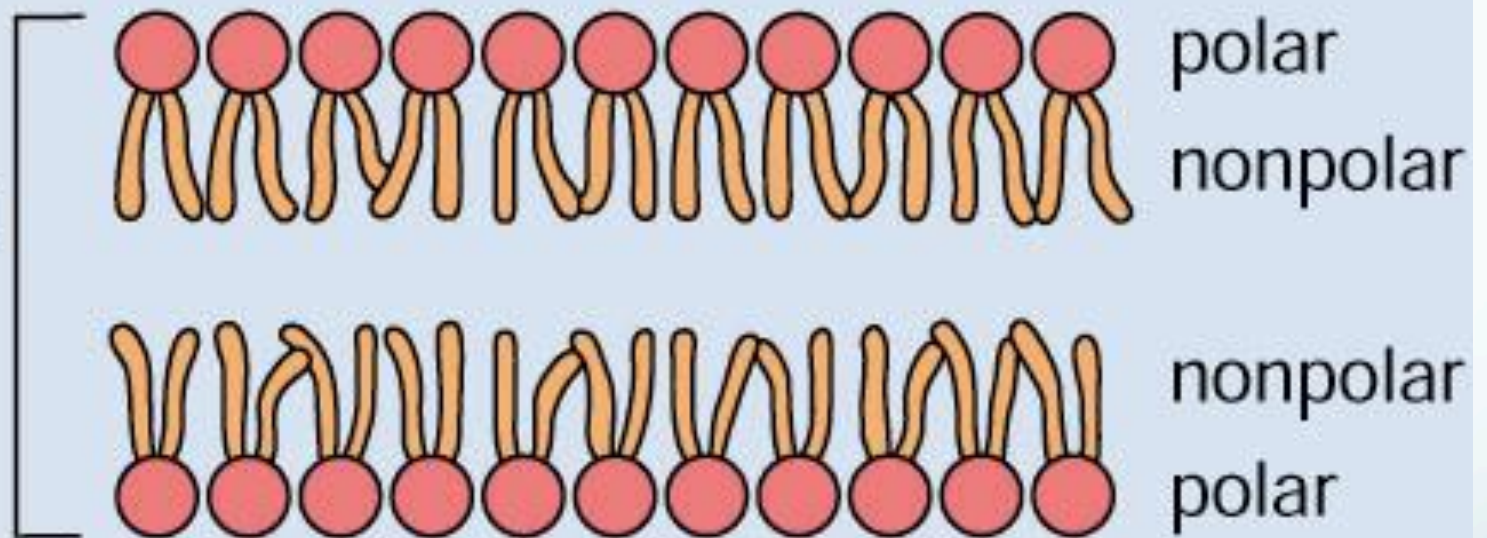
- **A-Phospholipids:**
- Are the most abundant lipids in most membranes, its an amphopathic molecule (meaning it has both hydrophilic & hydrophobic regions)



Phospholipid bilayer

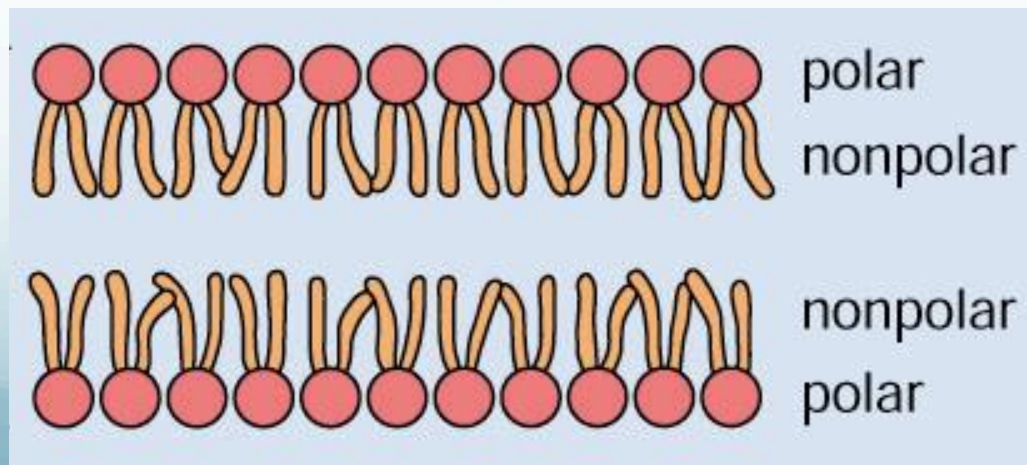


lipid
layer



- **A- Phospholipids**

The phospholipids in the plasma membrane are arranged in two layers, called a phospholipid bilayer, each phospholipid molecule has a head and two tails. The head “loves” water (hydrophilic) and the tails “hate” water (hydrophobic). The water-hating tails are on the interior of the membrane, whereas the water-loving heads point outwards, toward either the cytoplasm or the fluid that surrounds the cell.

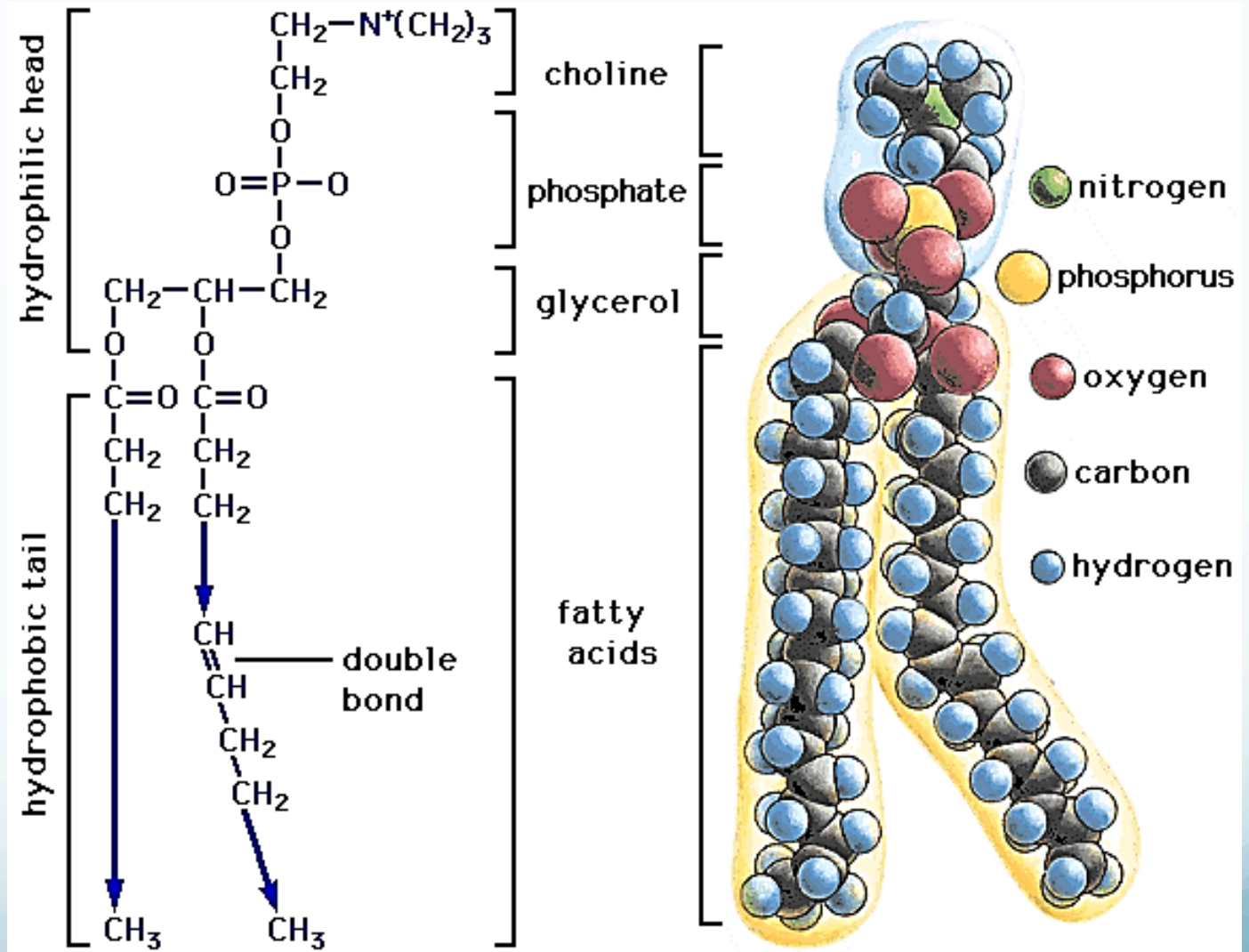


Phospholipid (Phosphatidylcholine)



Hydrophilic head

Hydrophobic tail

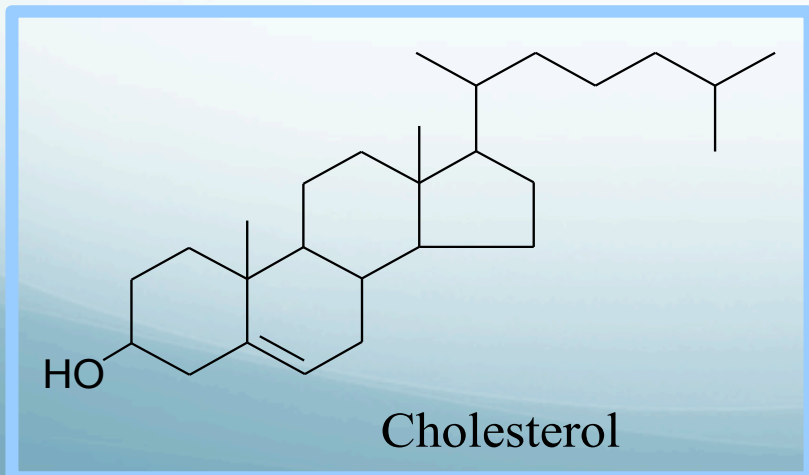


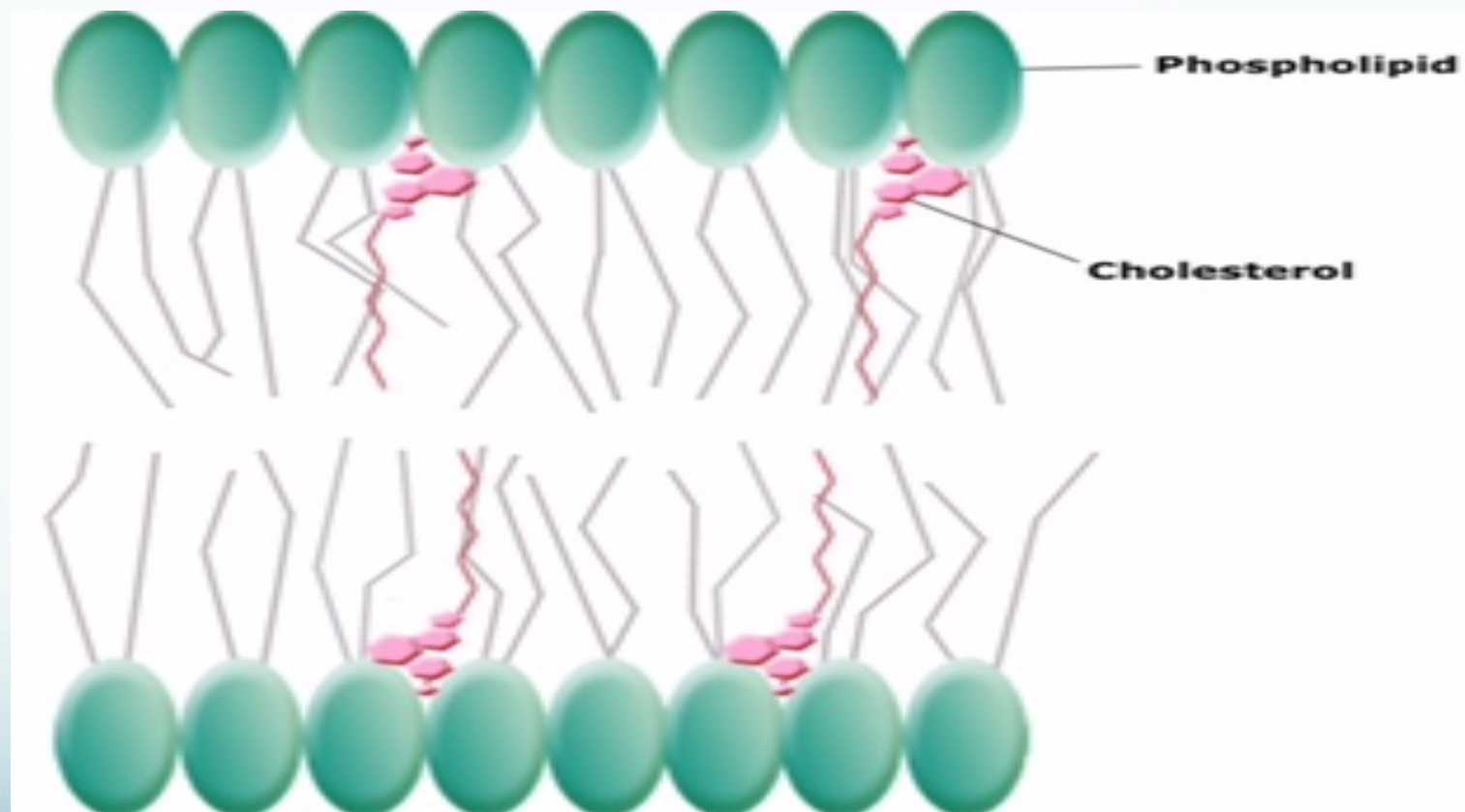
b-Cholesterol

Cholesterol inserts into bilayer membranes with its **hydroxyl group** oriented toward the aqueous phase & its hydrophobic ring system adjacent to fatty acid chains of phospholipids.

The OH group of cholesterol forms hydrogen bonds with polar phospholipid head groups.

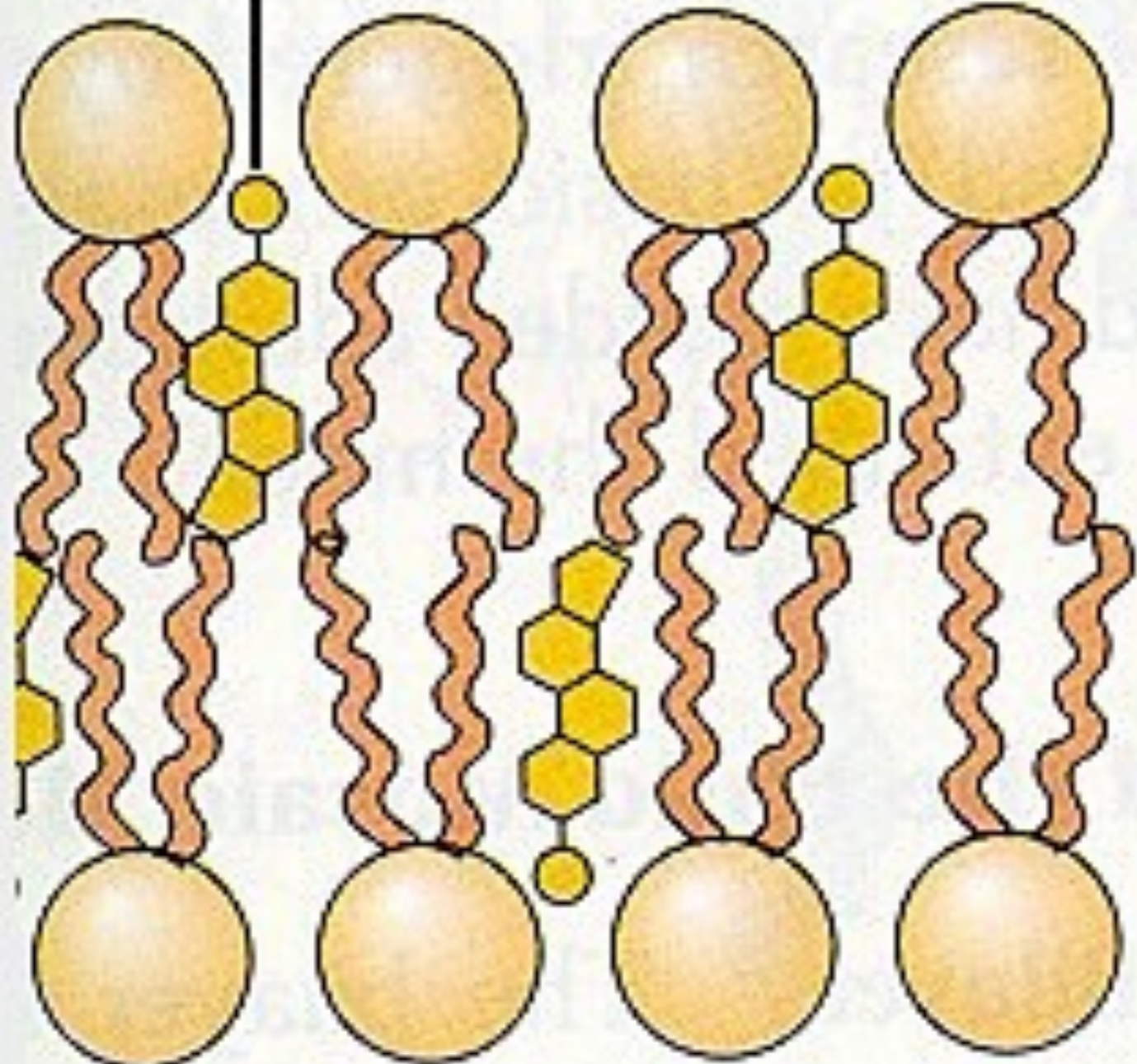
Cholesterol stiffens and strengthens the membrane, helping to regulate fluidity.





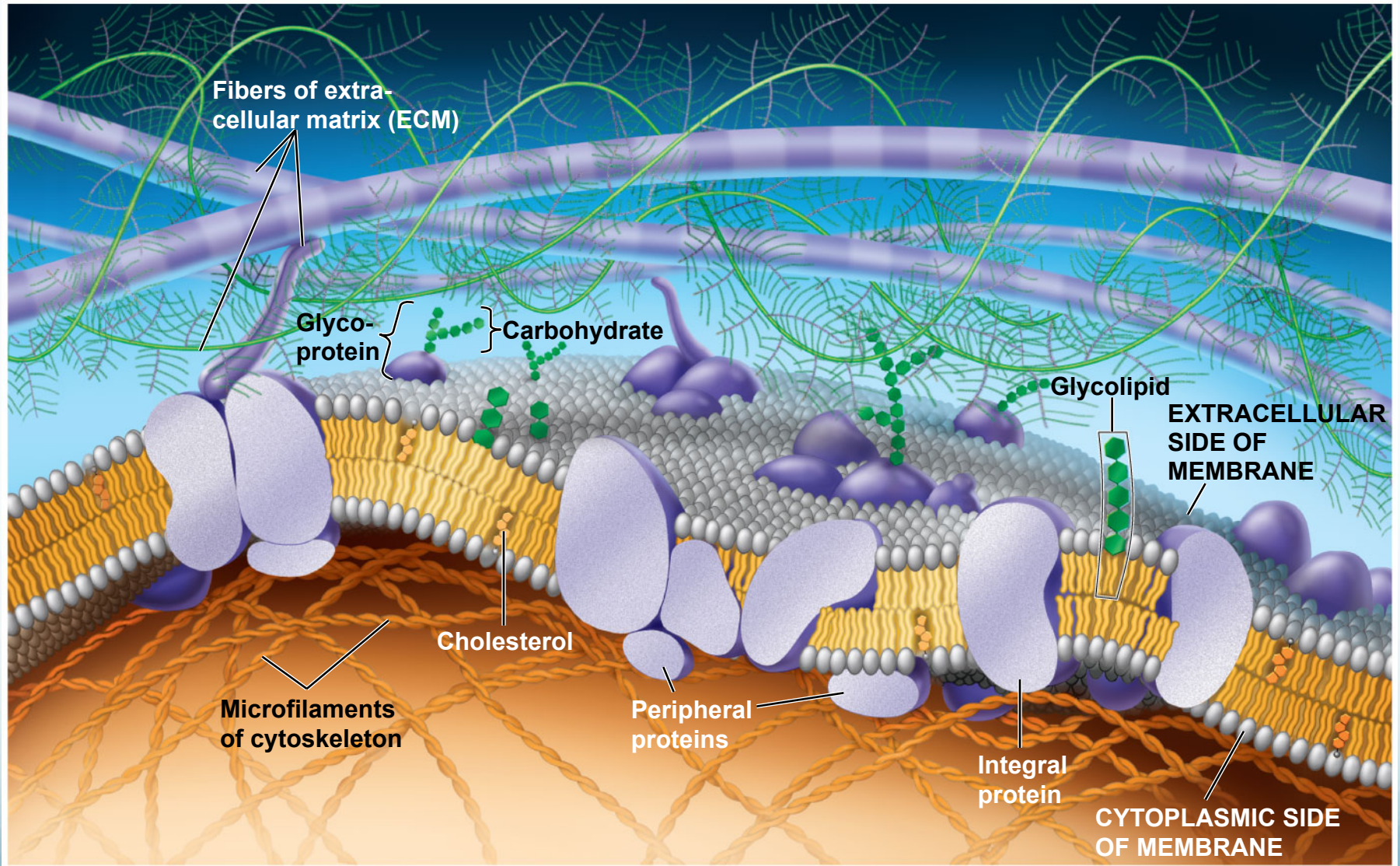
- Steroid cholesterol which is wedged between phospholipids effects membrane fluidity:
- A- at warm temperature it makes membrane less fluid by restraining the movement of phospholipids.
- B- at low temperature the membrane remains fluid because cholesterol hinders the close packing of the phospholipids.

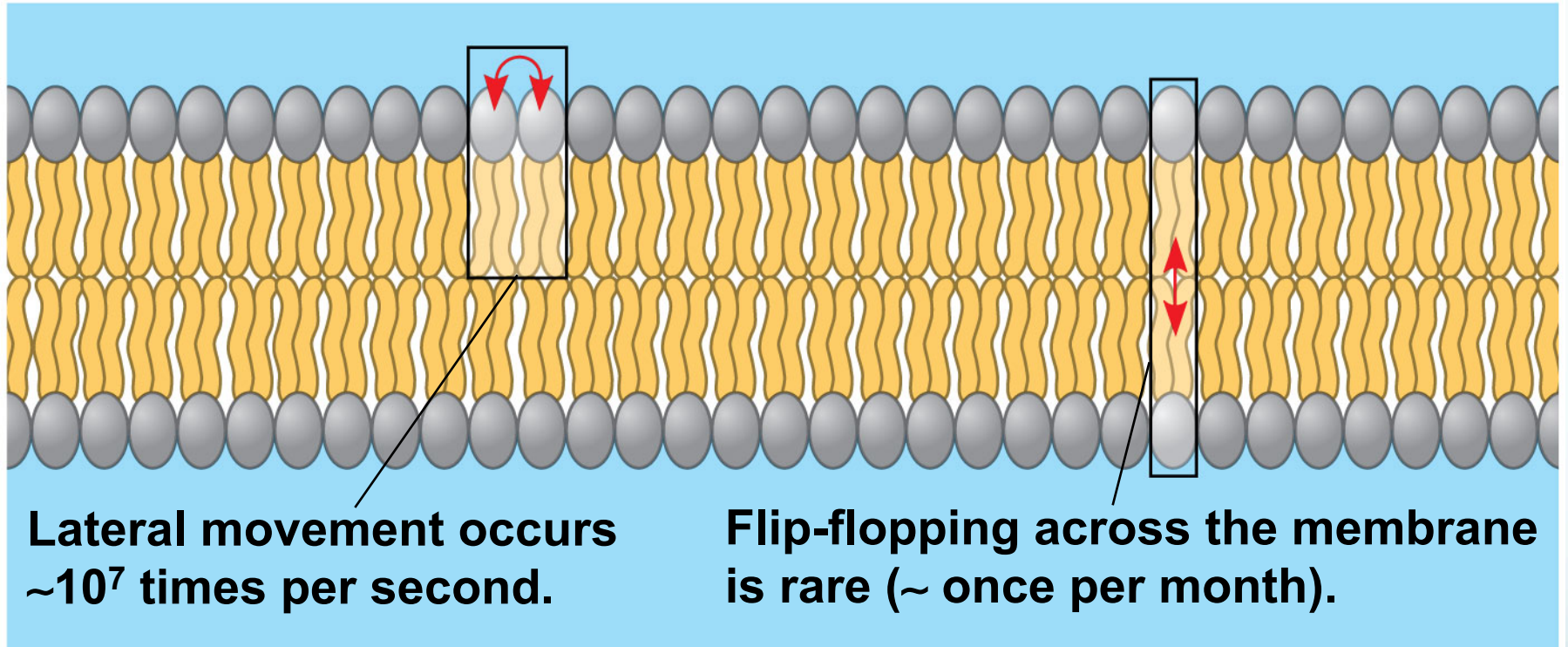
cholesterol



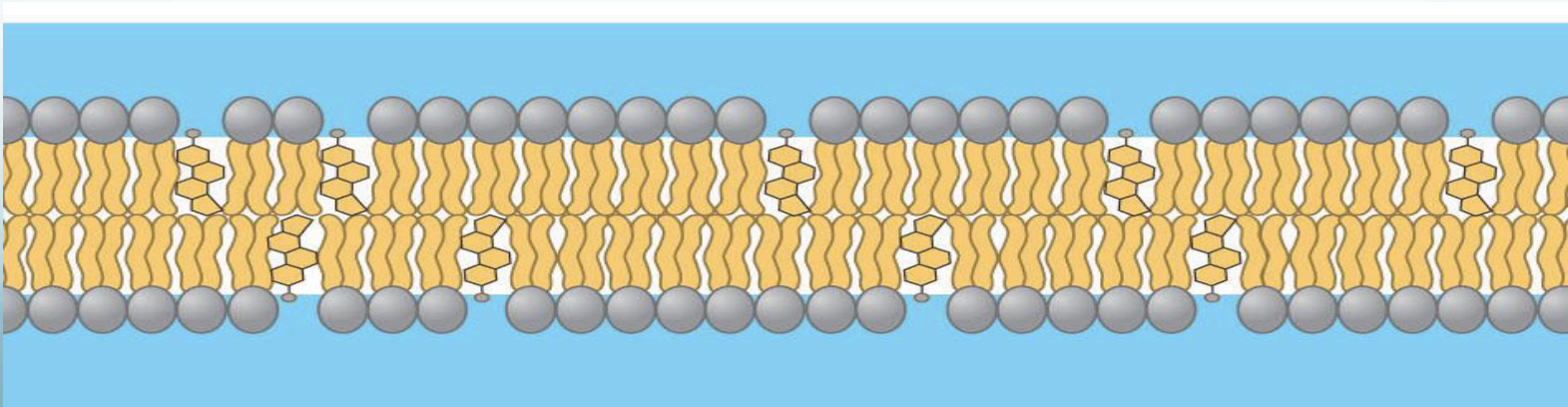
The Fluidity of Membranes

- Phospholipids in the plasma membrane can move within the bilayer
- Most of the lipids, and some proteins, drift laterally
- Rarely does a molecule flip-flop transversely across the membrane

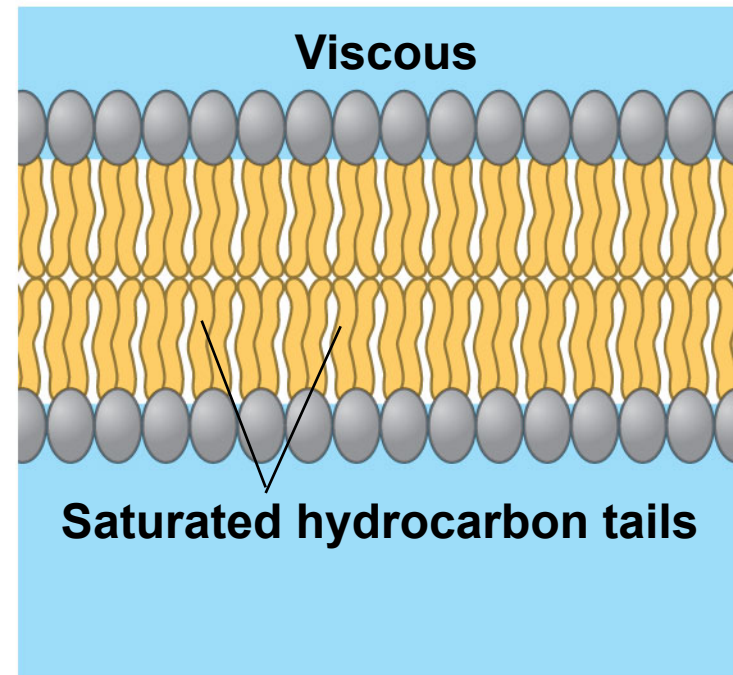
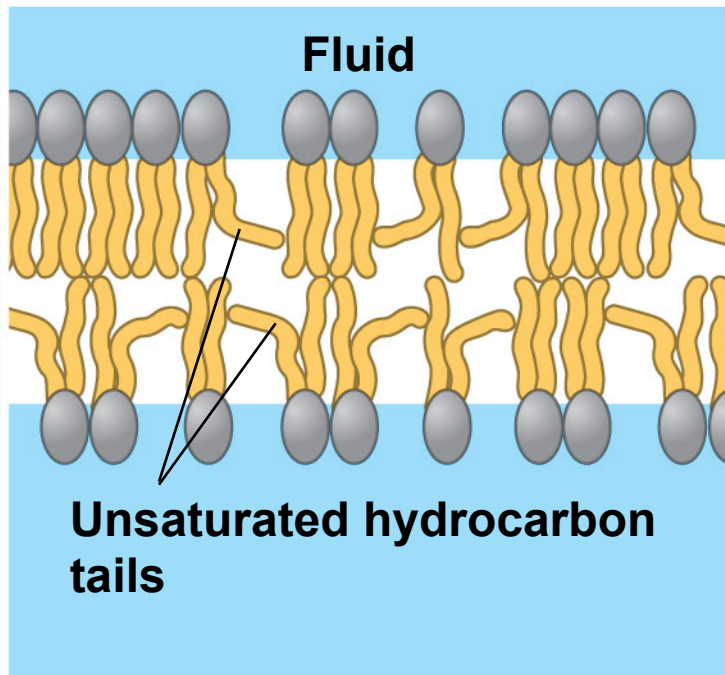




- The steroid cholesterol has different effects on membrane fluidity at different temperatures
- At warm temperatures (such as 37°C), cholesterol restrains movement of phospholipids
- At cool temperatures, it maintains fluidity by preventing tight packing

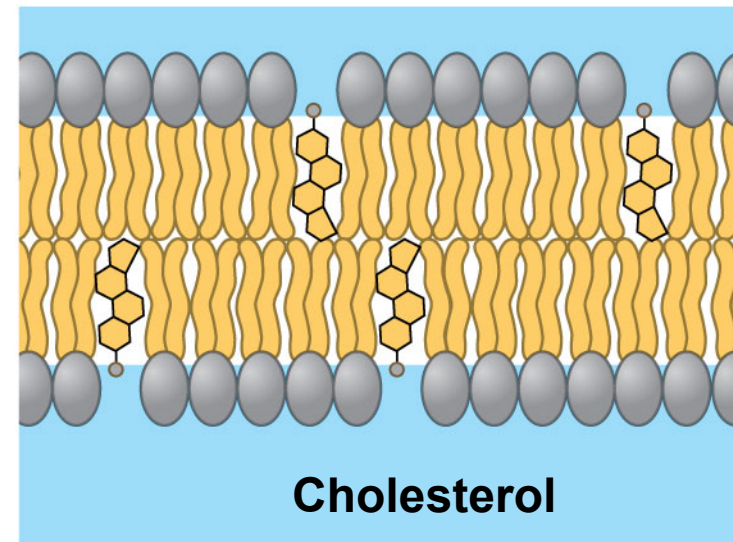


- As temperatures cool, membranes switch from a fluid state to a solid state
- The temperature at which a membrane solidifies depends on the types of lipids
- Membranes rich in unsaturated fatty acids are more fluid than those rich in saturated fatty acids
- Membranes must be fluid to work properly; they are usually about as fluid as salad oil

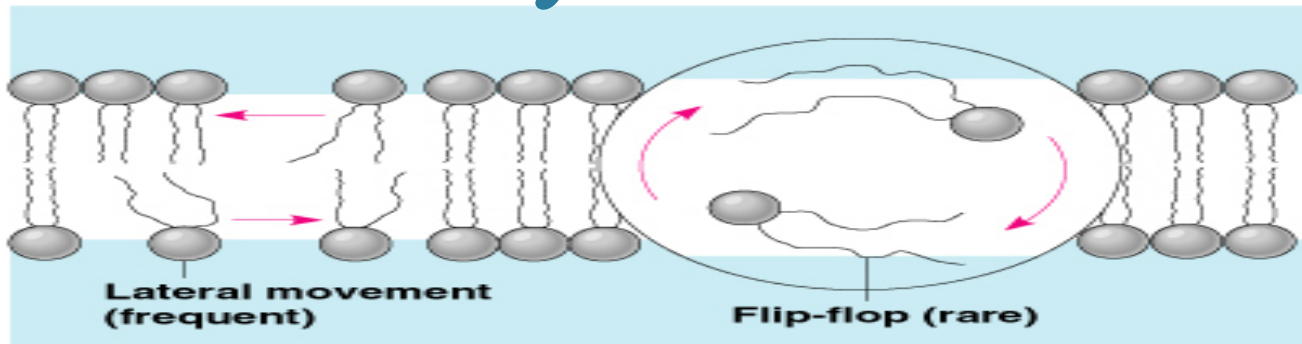


(a) Unsaturated versus saturated hydrocarbon tails

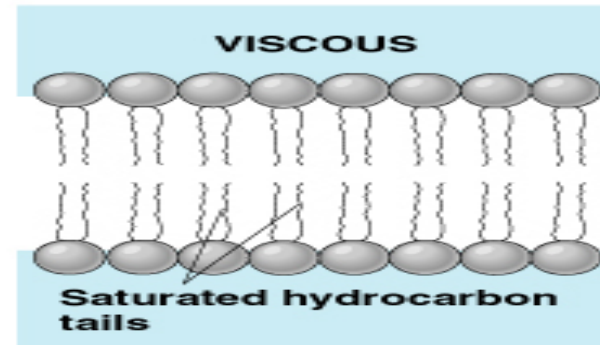
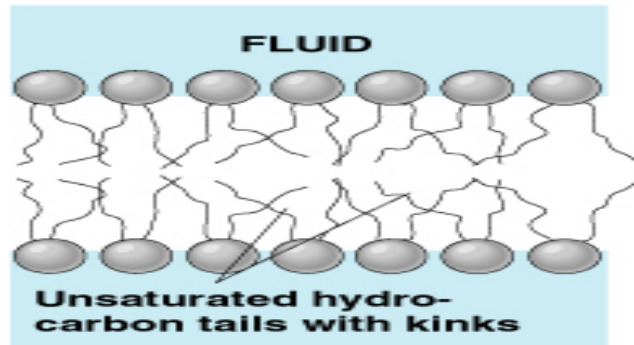
(b) Cholesterol within the animal cell membrane



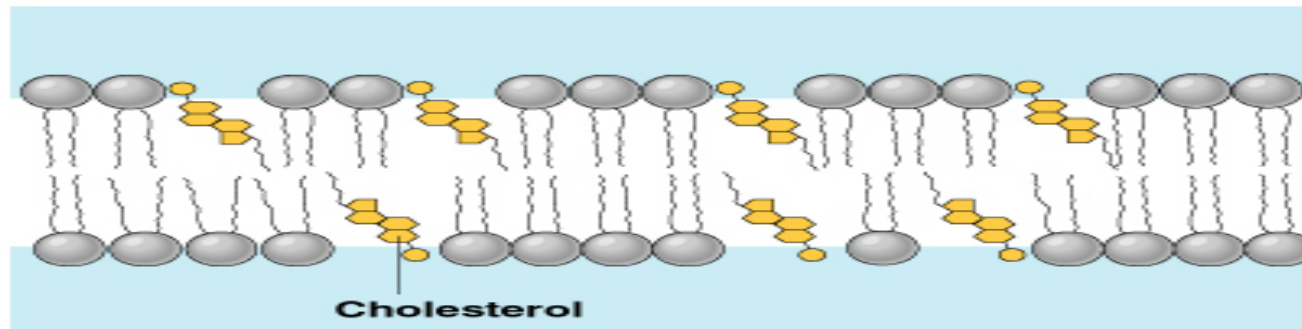
The Fluidity of Membranes



(a) Movement of phospholipids



(b) Membrane fluidity



(c) Cholesterol within the membrane

Evolution of Differences in Membrane Lipid Composition

- Variations in lipid composition of cell membranes of many species appear to be adaptations to specific environmental conditions
- Ability to change the lipid compositions in response to temperature changes has evolved in organisms that live where temperatures vary

Membrane Proteins and Their Functions

- A membrane is a collage of different proteins, often grouped together, embedded in the fluid matrix of the lipid bilayer
- Proteins determine most of the membrane's specific functions

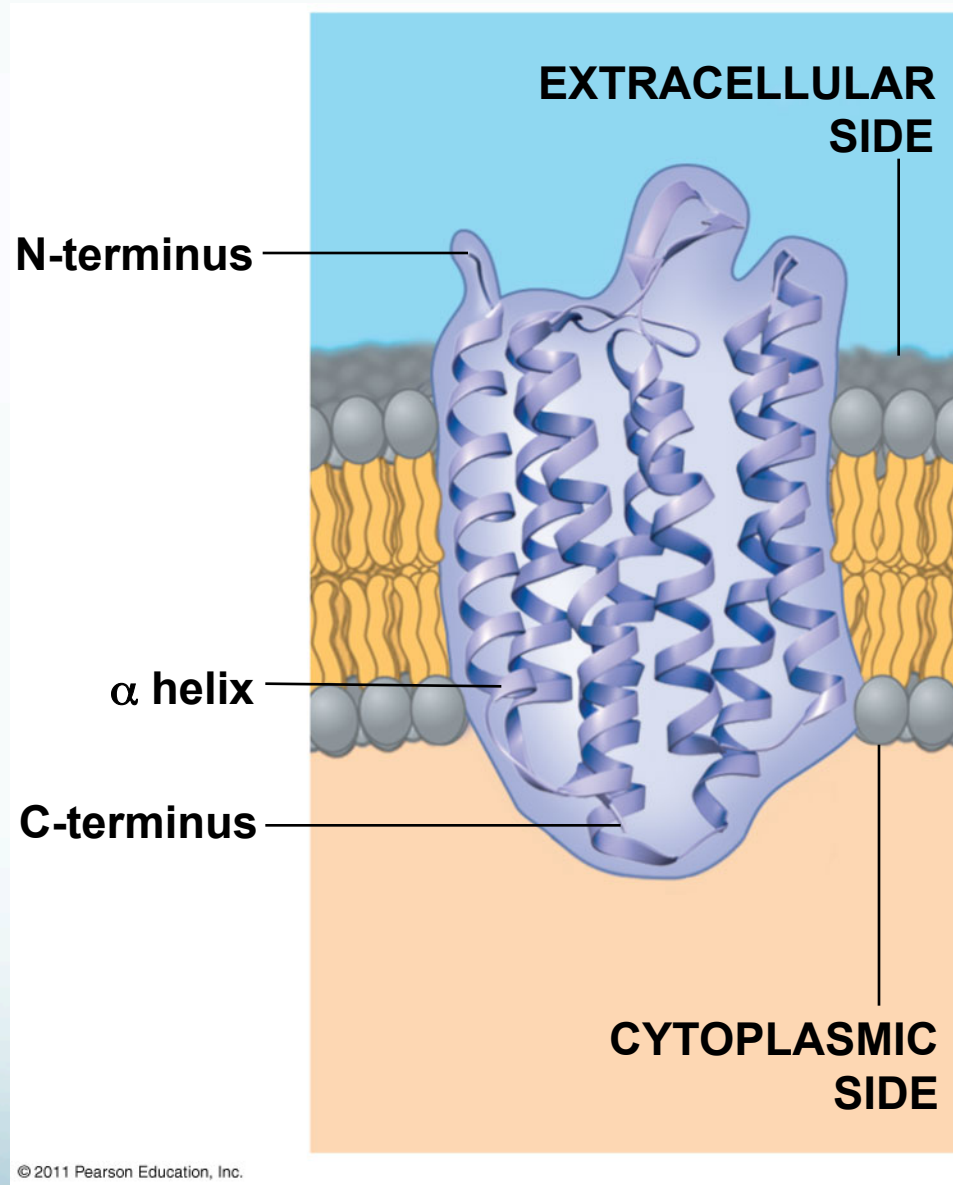
- **Peripheral proteins** are bound to the surface of the membrane
- **Integral proteins** penetrate the hydrophobic core
- Integral proteins that span the membrane are called transmembrane proteins
- The hydrophobic regions of an integral protein consist of one or more stretches of nonpolar amino acids, often coiled into alpha helices

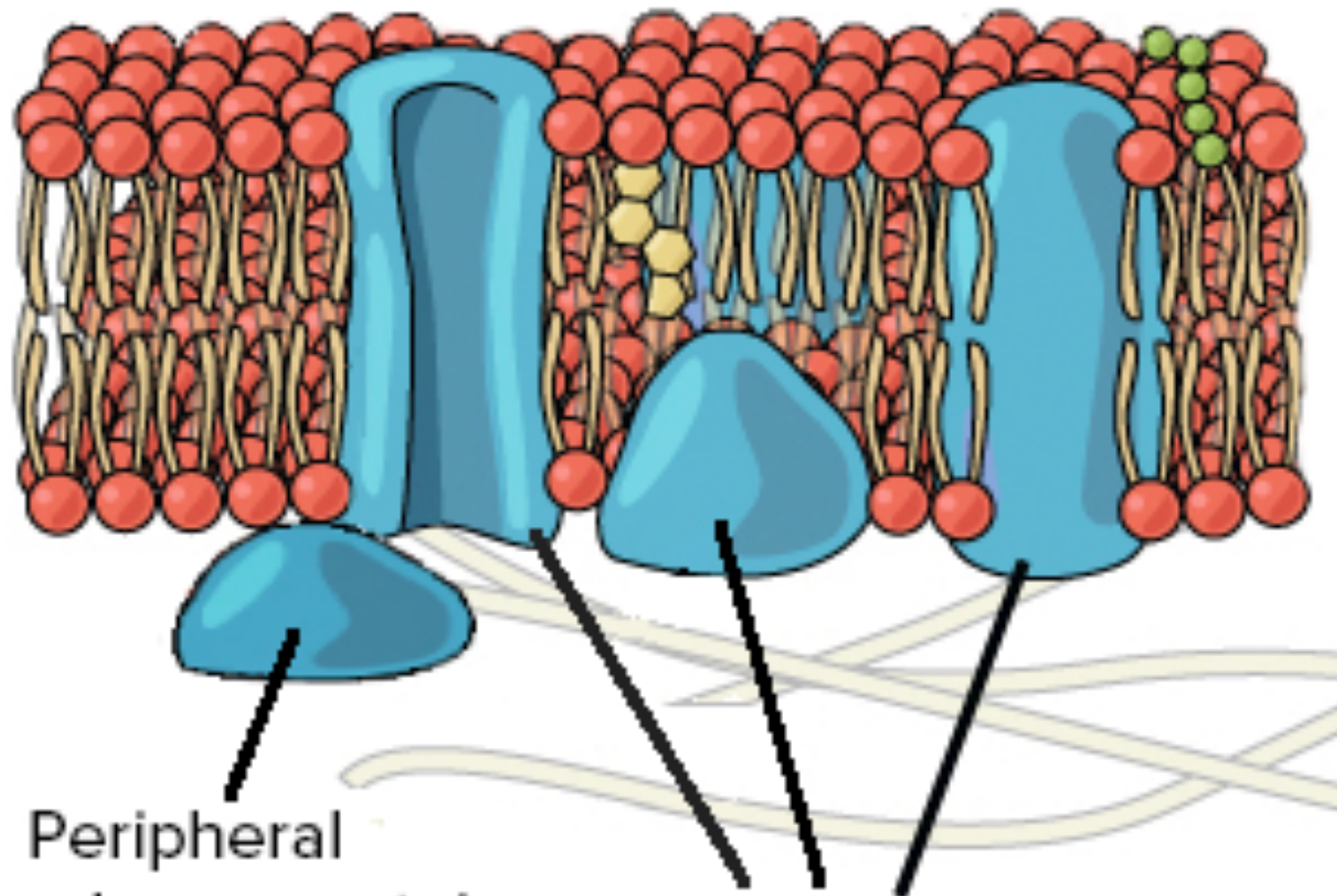
2-Proteins

- **A-Integral proteins:**

- Are inserted into the membrane & penetrate far enough in the phospholipids hydrophobic region.

This protein have important functions, they act as pumps for transport certain ions such as (Na^+) actively across membranes, also they act as channels which allow passage of small ions, also some proteins act receptors such as hormonal stimulation & antibody reaction.





Peripheral
membrane protein

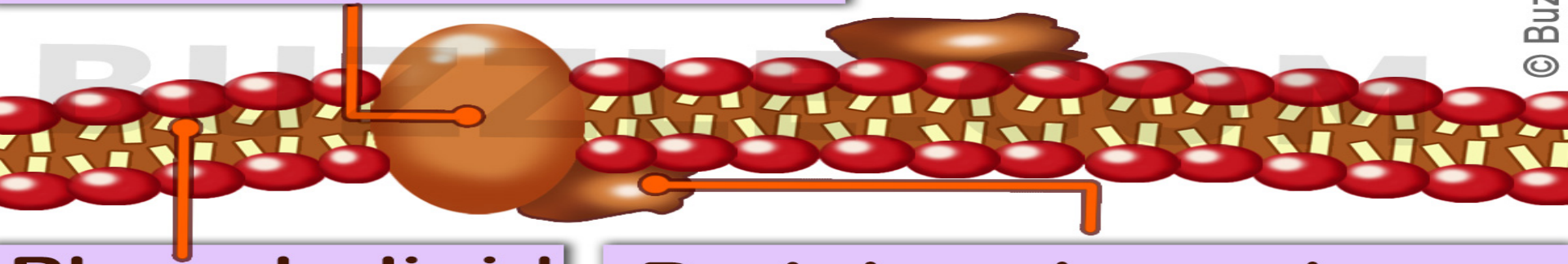
Integral membrane proteins

2-Proteins

- **B-Peripheral protein:**
- Are not embedded in the lipid bi layer at all, they are appendages attached to the surface of the membrane.

Cell Membrane

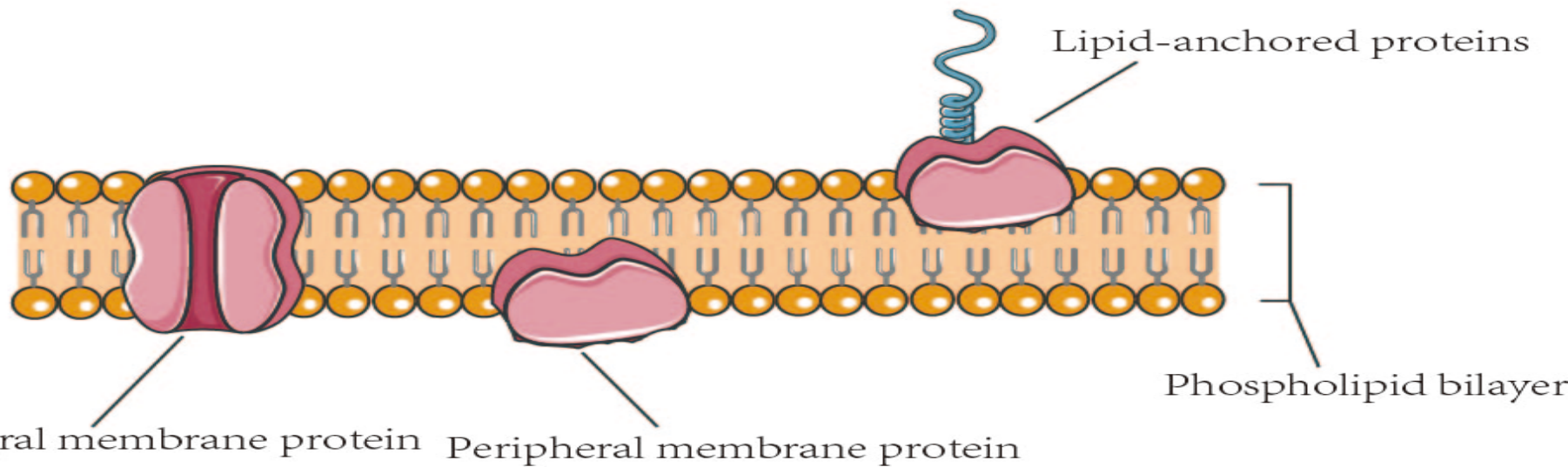
Integral membrane protein



Phospholipid bilayer

Peripheral membrane protein

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Lipid-anchored proteins

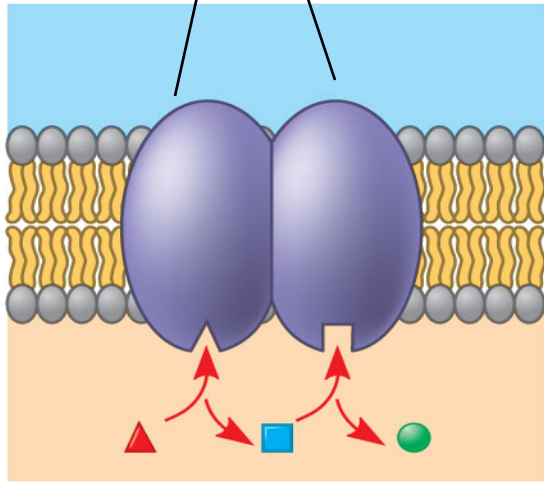
Phospholipid bilayer

Integral membrane protein

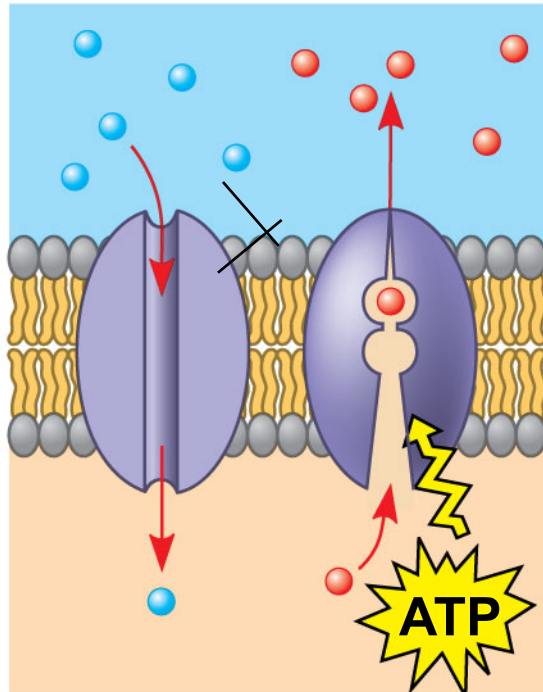
Peripheral membrane protein

- Six major functions of membrane proteins
 - Enzymatic activity
 - Transport
 - Signal transduction(receptors)
 - Cell-cell recognition
 - Intercellular joining
 - Attachment to the cytoskeleton and extracellular matrix (ECM)

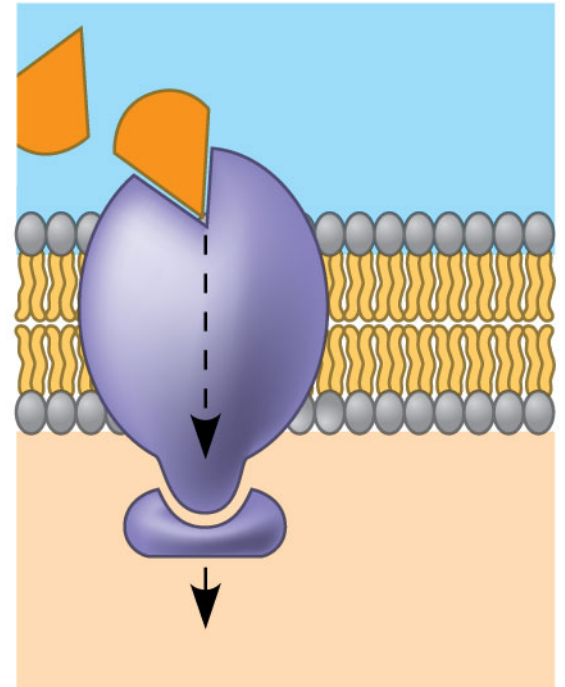
Enzymes



(a) Enzymatic activity



(b) Transport



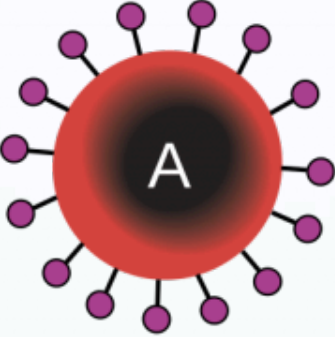
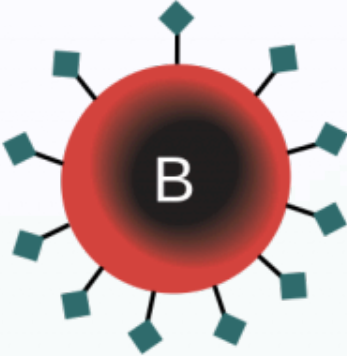
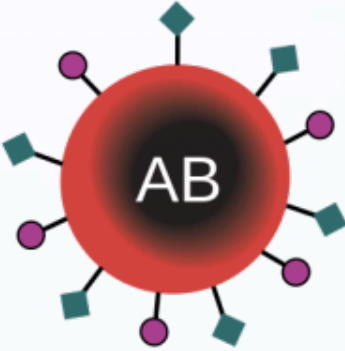
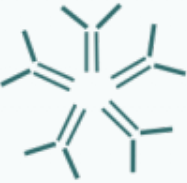
Signal transduction
(c) Signal transduction

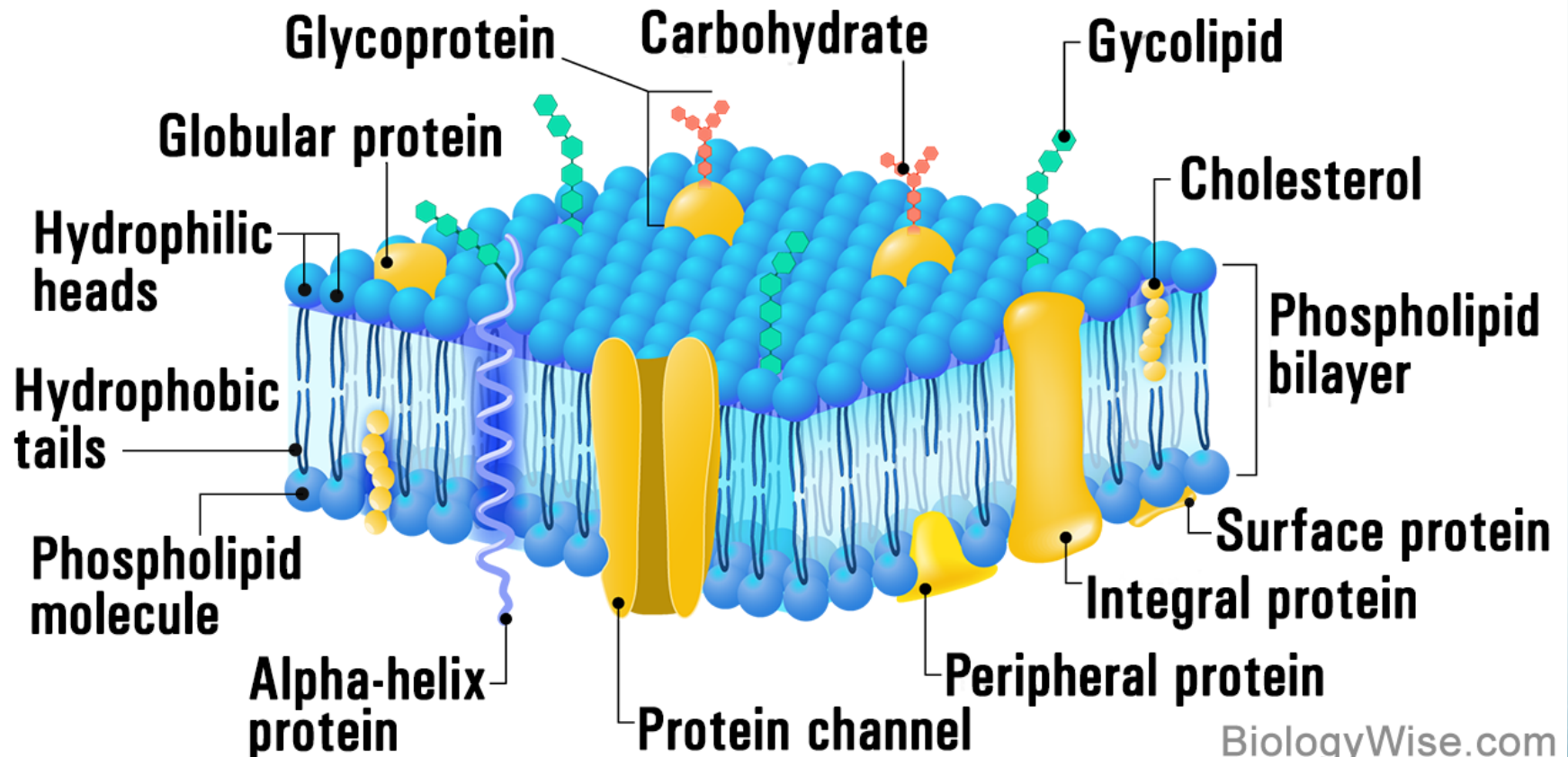
3- Carbohydrates

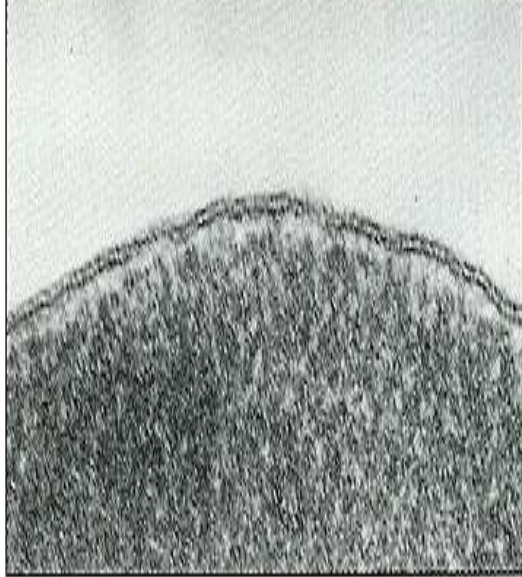
- Carbohydrates present in the plasma membrane as short sometimes branched chains of sugars attached either to exterior peripheral proteins (forming glycoproteins) or to the polar ends of phospholipid molecules in the outer lipid layer (forming glycolipids).
- Carbohydrate chains may consist of 2-60 monosaccharide units and can be either straight or branched.

3- Carbohydrate

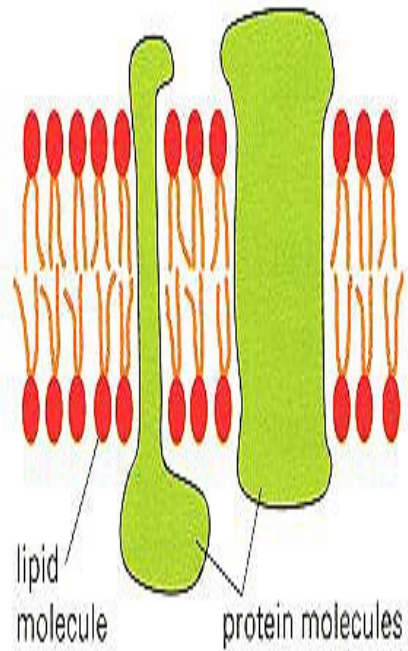
- Membrane carbohydrates perform **two main functions**:
- 1-participate in cell recognition and adhesion, either cell-cell signaling or cell-pathogen interactions
- 2-they have a structural role.
- -Blood groups are determined by cell surface carbohydrates of erythrocytes
- The carbohydrates on the external side of the plasma membrane vary from species to species and even from one cell type to another in a single individual. For example, the four human blood groups designated (A, B, AB, and O) reflects variations in the oligosaccharides on the surface of blood cell.

	Group A	Group B	Group AB	Group O
Red blood cell type				
Antibodies in Plasma	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens in Red Blood Cell	 A antigen	 B antigen	 A and B antigens	None

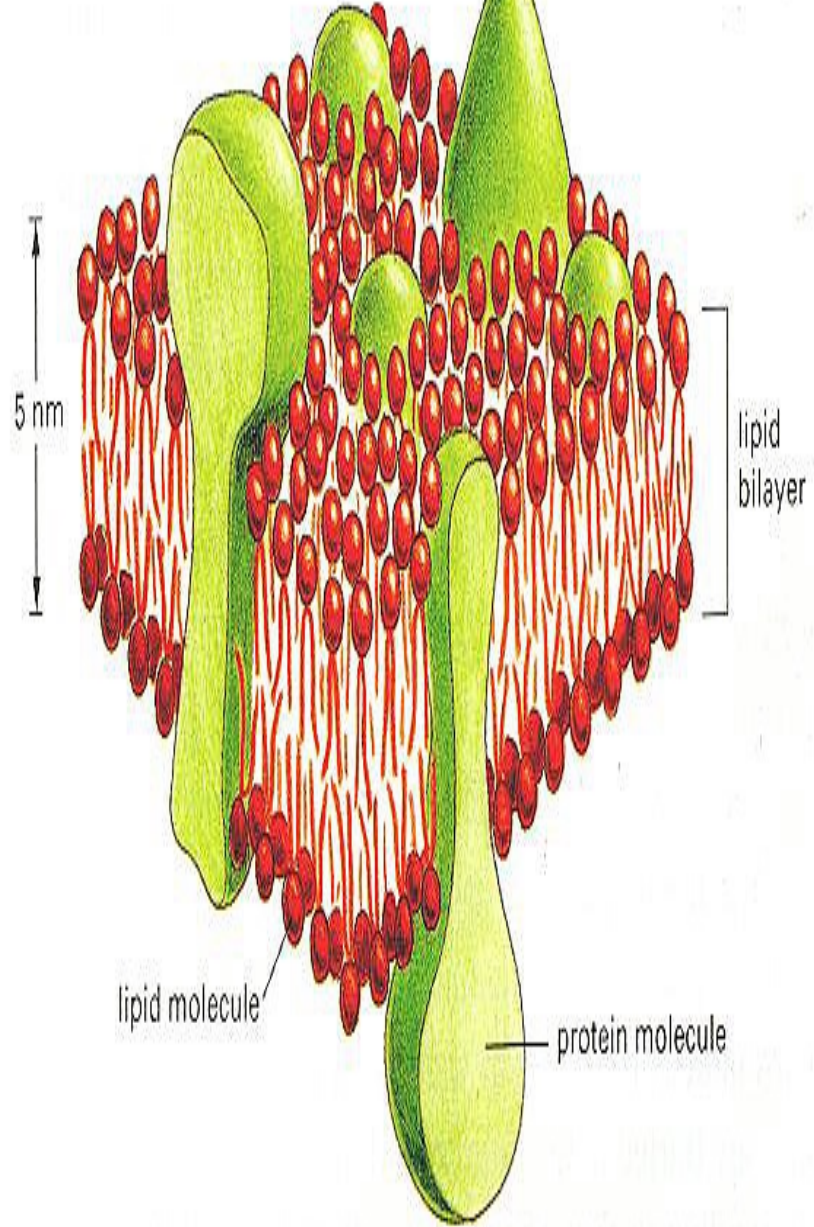




(A)



(B)



(C)

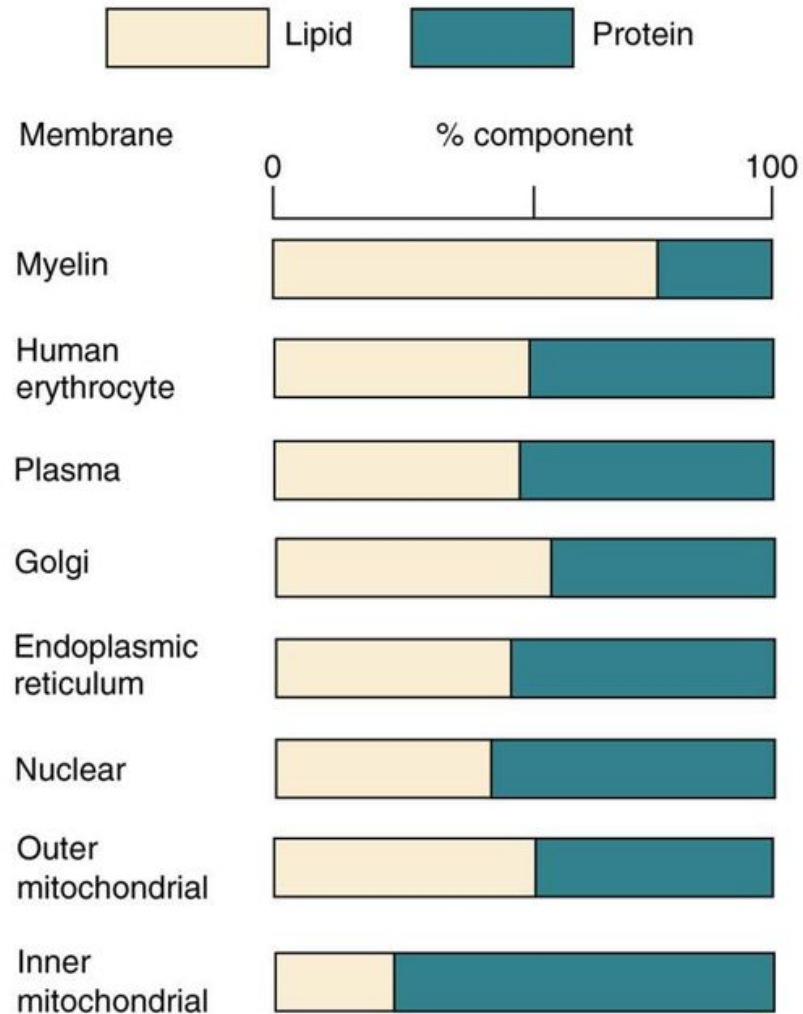


Figure 12.2 Percentage of lipid and protein in various cellular membranes.

<i>Component</i>	<i>Location</i>
Phospholipids	Main fabric of the membrane
Cholesterol	Tucked between the hydrophobic tails of the membrane phospholipids
Integral proteins	Embedded in the phospholipid bilayer; may or may not extend through both layers
Peripheral proteins	On the inner or outer surface of the phospholipid bilayer, but not embedded in its hydrophobic core
Carbohydrates	Attached to proteins or lipids on the extracellular side of the membrane (forming glycoproteins and glycolipids)

